Stakeholder Discovery and Software Engineering Research

COMP-7431: Advanced Software Engineering | Week 2
Software Architect · IBM Design Thinking · AI Periodic Table

Investigate the work before designing the system.

Monday Morning: Reconnect the Week 1 Map

WHAT WE CARRY FORWARD

1. The Software Architect connects people, decisions, and technology
2. IBM Design Thinking begins with human problems
3. The AI Periodic Table is a map of building blocks
4. The repository and AI Use Log preserve the engineering trail

TODAY'S SHIFT

5. Choose one uncertain situation
6. Identify everyone affected
7. Study the current journey
8. Gather and curate evidence
9. Frame a need without locking in the solution

Monday's first task is not "open the IDE." It is "reopen the map and decide what we still need to learn."

9:15 AM: The CampusConnect Assignment Arrives

THE SPONSOR'S REQUEST

"Students ask the same questions. Build an AI chatbot."

WHAT THE TEAM HEARS BEFORE ASKING

- Students sometimes contact the wrong office
- Staff repeat explanations and redirect requests
- Campus pages use overlapping or conflicting language
- Some questions carry financial, academic, or safety consequences

The request names a product. The Software Architect must discover the work, the evidence, and the risk behind it.

The Architect’s First Deliverable Is Better Understanding

Before choosing technology, the Software Architect helps the team:

- Name the decision that discovery must inform
- Identify users, operators, owners, and people who bear risk
- Observe how work really moves across pages, forms, and people
- Gather examples, artifacts, and operational evidence
- Separate facts, assumptions, unknowns, constraints, and risks
- Preserve contradictions instead of forcing early agreement

A useful early architecture artifact is a list of decisions that cannot yet be made responsibly.

Three Lenses Guide the Same Beginning

SOFTWARE ARCHITECT

- Frame the decision
- Map stakeholders and system context
- Expose constraints and risk
- Delay irreversible choices

IBM DESIGN THINKING

- Understand the user
- Map the as-is experience
- Find evidence-backed opportunities
- Invite correction

AI PERIODIC TABLE

- Gather relevant data
- Curate and label it
- Structure knowledge
- Validate provenance and quality

The methods differ, but all three protect the team from building intelligence on top of a false picture.

Week 2 Builds the Evidence Week 3 Will Need

By the end of this lecture, you should be able to:

- Explain why stakeholder discovery is architecture work
- Identify primary, operational, governance, and affected stakeholders
- Prepare ethical interviews and observations
- Map an as-is workflow and its exceptions
- Create an evidence ledger and empathy brief
- Use AI to cluster de-identified notes with human review
- Write a need and problem statement that keeps alternatives open
- Carry one important unanswered question into the lab

A Requested Solution Is Not Yet a Problem Definition

REQUESTED SOLUTION

"Build an AI chatbot for student questions."

INVESTIGATED SITUATION

- Students cannot consistently identify the correct service
- Similar services use overlapping language
- Information is scattered and sometimes contradictory
- Misrouted requests create delay and repeated explanations
- High-consequence guidance needs reliable human escalation

Chat is one possible interface. The problem may involve taxonomy, content ownership, search, routing, or workflow.

CampusConnect Begins with Evidence and Unknowns

WHAT WE CAN SAY TODAY

Some students reach the wrong service and repeat their story.

WHAT WE MUST LEARN NEXT

- Which student groups experience the problem most often
- Where the current journey breaks down
- Which labels, pages, forms, or handoffs cause confusion
- How staff repair a misrouted request
- Who owns the accuracy of guidance
- Which situations require a human immediately

A strong discovery question names the decision it will improve—not merely the information it hopes to collect.

Discovery Unlocks Specific Architecture Decisions

The architect lists decisions that are premature:

- Interface: chat, search, guided form, voice, or hybrid?
- Knowledge: which sources are authoritative and who updates them?
- Identity: anonymous guidance, authenticated service, or both?
- Handoff: when does a person take over, and what context follows?
- Data: what is collected, retained, minimized, or prohibited?
- Boundary: which student journeys belong in the first release?
- Risk: where must the system refuse, qualify, or escalate?

Mini example: do not choose a vector database until you know what content is trustworthy enough to retrieve.

Stakeholders Use, Operate, Govern, and Bear Risk

Stakeholder role	CampusConnect example	Evidence they contribute
Primary experience	Students seeking help	Goals, language, barriers, trust
Operational work	Advisors and departments	Routing, exceptions, workload
System operation	IT, help desk, content owners	Support, failures, maintenance
Governance and risk	Security, privacy, accessibility	Controls, obligations, boundaries
Authority or indirect	Sponsor, leaders, peer mentors	Priorities, funding, secondary impact

Stakeholder status comes from impact or responsibility—not job title alone.

A Stakeholder Map Prevents Convenient Blind Spots

PRIMARY EXPERIENCE

- Current students
- Prospective students
- Multilingual users
- Students using assistive technology

OPERATIONAL WORK

- Advisors
- Service departments
- Help desk
- Content owners

AUTHORITY AND RISK

- Project sponsor
- IT and security
- Accessibility staff
- Privacy or legal review

If a group can be helped, burdened, excluded, or made responsible, it belongs on the map.

Influence and Impact Shape How We Engage

Classify people by decision influence and degree affected:

- High influence + high impact: collaborate frequently
- High influence + lower impact: clarify decisions and constraints
- Lower influence + high impact: protect representation and access
- Lower influence + lower impact: keep informed and watch for change
- Revisit the map when scope, risk, or ownership changes

Co-design → **Interview** → **Review** → **Keep informed**

The Sponsor’s View Is Essential—but Incomplete

Sponsor statement	Software Architect’s investigation
“We need a chatbot.”	Is the dominant failure search, content, routing, or handoff?
“Students want it.”	Which students, in which situations, with what evidence?
“Use our current pages.”	Are the pages accurate, consistent, accessible, and owned?
“Launch this term.”	Which journey is safe and feasible within that boundary?
“Success means adoption.”	Do misrouting, delay, completion, and trust improve?

Questioning a request is not resistance. It is how the architect protects the intended outcome.

Conflicting Needs Are Normal Engineering Inputs

STUDENTS

- Fast next step
- Plain language
- Accessible interaction
- Confidence in the guidance

DEPARTMENTS

- Accurate routing
- Fewer repeated contacts
- Manageable content updates
- Clear ownership

IT AND GOVERNANCE

- Supportable design
- Privacy and access control
- Reliable escalation
- Traceable change

Discovery makes tensions visible before architecture silently resolves them.

Research Ethics Starts Before the First Question

A responsible discovery protocol makes participation safe:

- Explain the purpose, activity, and expected use of notes
- Make participation voluntary and allow people to stop
- Collect only information the decision genuinely needs
- Avoid grades, services, or authority influencing participation
- Ask permission before recording; offer note-taking instead
- De-identify notes and separate identity from findings
- Avoid requesting passwords, records, diagnoses, or private data

Ask what made the financial-aid journey confusing; never request a private account balance.

A Discovery Plan Connects Each Activity to a Decision

A practical plan identifies:

- The architecture or product decision the evidence must inform
- Stakeholder roles and a purposeful, feasible sample
- Interview, observation, document, and data methods
- A short protocol with neutral questions and probes
- Consent, privacy, note-taking, and recording boundaries
- Where artifacts and source IDs will be stored
- Who synthesizes, who checks, and when the team stops

Discovery is deliberately small: enough evidence to make the next decision responsibly.

Different Methods Reveal Different Evidence

Method	Best for revealing	CampusConnect mini example
Interview	Goals, stories, meaning, exceptions	Recent misrouting experience
Observation	Actual steps, workarounds, waiting	Student attempts a real search
Document review	Rules, ownership, official language	Service pages and policies
Operational data	Frequency, patterns, concentration	Tickets, search terms, reroutes
Concept review	Understanding and early reactions	Directory versus guided chat

When ambiguity or consequence is high, use more than one method.

Strong Interviews Collect Stories, Not Feature Votes

A useful interview is specific, neutral, and grounded in a recent experience:

- Ask about the last real occurrence—not an imagined future
- Begin open; probe details after the story emerges
- Ask what happened before, during, and after the breakdown
- Separate observed behavior from stated preference
- Request examples, artifacts, and exceptions when appropriate
- Avoid selling the proposed solution during discovery
- Summarize what you heard and invite correction

The stakeholder supplies lived experience; the architect supplies disciplined listening.

Neutral Questions Produce More Useful Evidence

Weak or leading question	Stronger discovery question
Would you use a chatbot?	Tell me about the last time you needed campus help.
Do you like the website?	Show me how you tried to find the answer.
What features should we add?	What did you try, and what happened next?
Is the routing confusing?	How did you choose which department to contact?
Would AI make it easier?	What would make guidance reliable enough to act on?

Good questions reduce pressure to agree and create room for unexpected evidence.

A Short Interview Reveals Language, Workflow, and Risk

OPEN QUESTION

"Tell me about the last time you were unsure which campus office to contact."

STAKEHOLDER EVIDENCE

"I searched 'drop class,' found registrar and advising pages, then texted a friend because the deadlines looked different."

PROBE AND ARCHITECTURE LEARNING

Probe: "What happened after you texted your friend?"

Learning: the dominant problem may be conflicting terminology and content ownership—not the absence of chat.

Observation Makes Invisible Work Visible

During observation, record what actually happens:

- Steps, decisions, tools, pages, forms, and handoffs
- Waiting, interruption, rework, and repeated entry
- Workarounds that compensate for the official process
- Exceptions that do not fit the happy path
- Information copied between systems or people
- Moments where confidence rises or collapses
- The point where the participant gives up or asks for help

Three tabs and a text to a friend mean “page loaded” did not equal “journey succeeded.”

The As-Is Workflow Shows Handoffs and Rework

Search campus site

↓

Open two service pages

↓

Choose a department

↓

Submit the request

↓

Wait for a response

↓

Repeat the story elsewhere

EVIDENCE TO CAPTURE

- Time and waiting
- Decision points
- Handoffs
- Failed attempts
- Workarounds
- Responsible owner

An As-Is Scenario Map Connects Evidence and Consequences

STEPS AND TOUCHPOINTS

- Trigger and intended outcome
- Actions and decisions
- Pages, forms, systems, people
- Handoffs and waiting

EVIDENCE AND EXPERIENCE

- Quotes and observed behavior
- Confidence and uncertainty
- Effort, delay, and rework
- Exceptions and workarounds

BREAKDOWNS AND QUESTIONS

- Pain points and consequences
- Root-cause hypotheses
- Ownership gaps
- What the team must investigate next

A scenario map is credible only when its important claims can be traced to evidence.

Documents and Data Test the Interview Story

Review the artifacts that already shape the work:

- Service pages, policies, FAQs, and published deadlines
- Forms, required fields, and routing rules
- Search terms, page exits, and failed queries
- Tickets, reroutes, repeated contacts, and resolution time
- Content owners, update dates, and approval responsibilities
- Accessibility reports, incidents, and known exceptions
- Existing service-level expectations and escalation paths

Mini example: two pages with different deadlines indicate a governance problem—not merely a chatbot problem.

Triangulation Keeps Contradictions Visible

1. Interviews explain how people interpret the experience
2. Observation reveals actual behavior and workarounds
3. Documents reveal intended rules, language, and ownership
4. Operational data reveals frequency and concentration
5. The team compares convergence, contradiction, and missing evidence

Contradiction is a finding: reported ease can coexist with workarounds and repeated queries.

The AI Periodic Table Begins with a Knowledge Foundation

GATHER

- Interviews and observations
- Pages, policies, forms
- Search and ticket examples
- Exceptions and edge cases

CURATE

- De-identify sensitive details
- Remove duplicates
- Label source and date
- Separate evidence from inference

STRUCTURE AND VALIDATE

- Evidence ledger and source IDs
- Themes with counterexamples
- Ownership and provenance
- Quality checks before reuse

A RAG system, agent, or chatbot built later will inherit the strengths—and the weaknesses—of this foundation.

An Evidence Ledger Keeps Knowledge Honest

Status	CampusConnect example	Next action
Evidence	18 of 50 sampled tickets were rerouted	Segment by request type
Assumption	Students prefer conversation	Compare chat and search concepts
Constraint	No unnecessary record access	Minimize data and permissions
Unknown	No clear owner for answer accuracy	Resolve content governance
Risk	Incorrect financial-aid guidance	Require sources and escalation

An Empathy Brief Separates Evidence from Interpretation

OBSERVED OR REPORTED

- Exact quote or careful paraphrase
- Steps taken and tools used
- Workaround selected
- Delay, rework, or consequence
- Source note identifier

INFERRED—THEN VALIDATED

- Motivation or desired progress
- Confidence or trust concern
- Possible unstated need
- Emotional interpretation
- Alternative explanation

Empathy is disciplined perspective-taking: label inference, preserve evidence, and invite the person to correct you.

AI Organizes Evidence—but It Cannot Witness

Discovery activity	Appropriate AI assistance	Required human control
Note cleanup	Format de-identified notes	Consent and privacy review
Theme clustering	Group repeated patterns	Keep source IDs and dissent
Summarization	Draft concise findings	Verify consequential claims
Question design	Suggest neutral probes	Remove leading assumptions
Gap analysis	Flag missing roles or evidence	Choose the next method
Problem framing	Generate alternative wording	Trace claims to evidence
Artifact drafting	Format maps and ledgers	Never fabricate observations

A Good AI Prompt Preserves Traceability and Disagreement

SAFE INPUT

De-identified notes with source IDs such as INT-01, OBS-02, and DOC-03.

ANALYSIS PROMPT

Cluster the notes by repeated need, workflow breakdown, constraint, and contradiction. Quote only supplied text. Cite source IDs. Mark low-confidence themes and preserve counterexamples.

HUMAN REVIEW BEFORE ACCEPTANCE

- Verify every quote and source ID
- Keep outliers
- Separate evidence from inference
- Reject invented details
- Record revisions and AI use

Symptoms Point to Problems; They Do Not Explain Them

Visible symptom	Underlying condition to investigate
Students contact the wrong office	Overlapping names or unclear service taxonomy
Staff repeat the same explanation	Scattered content or weak self-service
A chatbot gives a wrong answer	Contradictory sources or missing ownership
Requests are abandoned	Unclear next step, delay, or inaccessible handoff
Peak periods overwhelm staff	Deadlines, batching, or avoidable rerouting

Fixing only the symptom can automate the same failure faster.

Root-Cause Questions Move Toward Actionable Conditions

CampusConnect example: students are repeatedly routed to the wrong office.

- Ask why each condition exists and cite evidence at every step
- Follow more than one causal branch when needed
- Stop where the cause becomes actionable within the project boundary
- Validate the chain with people, artifacts, or data
- Keep alternative explanations visible
- Do not confuse a plausible story with a demonstrated cause

Wrong office → **Different labels** → **Department-first search** → **No shared taxonomy**

A Needs Statement Describes Progress, Not Technology

A useful needs statement has three parts:

- When… describe the triggering situation
- I want to… describe the progress or action
- So I can… describe the meaningful outcome
- Keep the statement independent of a proposed product
- Use discovery evidence—not demographic stereotypes
- Compare several solutions against the same need

"When I am unsure who owns my issue, I want a reliable next step so I can act without repeating my story."

A Problem Statement Names the Barrier and Consequence

1. Stakeholder — who experiences the problem
2. Context — where and when it occurs
3. Desired progress — what the person is trying to accomplish
4. Barrier — what prevents that progress
5. Consequence — delay, risk, cost, rework, or exclusion
6. Desired outcome — the observable improvement that matters

[Stakeholder] cannot [desired progress] when [context] because [barrier], causing [consequence].

Evidence Keeps the Problem Open to Better Solutions

SOLUTION-LOCKED STATEMENT

"Students need an AI chatbot."

→

EVIDENCE-BACKED PROBLEM

Students seeking campus services cannot reliably identify the correct owner or next step when service language overlaps. They experience delay, repeated contacts, and risk from inconsistent guidance.

WHY THE SECOND IS STRONGER

- Names the affected group and situation
- Identifies an evidence-supported barrier
- States human and operational consequences
- Keeps search, workflow, content, human, and AI options open
- Points toward observable outcomes

Measure Better Outcomes—not More Features

Feature or activity measure	Stakeholder outcome measure
Launch a chatbot	Reduce avoidable misrouting
Count conversations	Increase successful next-step completion
Measure response speed	Reduce time to the correct service
Track sign-ups	Reach affected groups accessibly
Count generated answers	Improve accuracy and source traceability
Record AI confidence	Escalate uncertainty appropriately

A metric is useful only when it represents the intended stakeholder outcome.

Constraints and Edge Cases Reveal the Real Boundary

Discovery scenario or constraint	Architecture question it creates
Student uses a screen reader	Can the journey work by keyboard and assistive technology?
Student has limited English proficiency	How will terminology and language be tested?
Question concerns safety or health	When must the system escalate immediately?
User cannot authenticate	Which journeys may remain anonymous?
Source content is outdated	Who owns review, correction, and fallback?
No unnecessary record access	What data and permissions can be removed?
Department is overloaded	How will capacity, delay, and alternative routing appear?

Edge cases are not decorative examples; they expose requirements the average journey hides.

The Lab Produces a Traceable Discovery Package

PEOPLE AND CURRENT WORK

1. Stakeholder map
2. Ethical discovery protocol
3. Interview or observation notes
4. Empathy brief
5. As-is workflow or scenario map

EVIDENCE AND DIRECTION

6. Pain points and root-cause hypotheses
7. Evidence / assumption / unknown ledger
8. Initial needs and problem statement
9. Success measures and constraints
10. Engineering/AI Use Log

Every conclusion should show where it came from, what it means, and what remains uncertain.

Lab Preview: Investigate Before You Define

“Your team is not collecting decoration for a slide. It is building the evidence that Week 3 will trust.”

- Map everyone affected or responsible
- Ask for recent stories—not feature votes
- Observe the current workflow and exceptions
- Build empathy and as-is maps from evidence

- Label evidence, assumptions, unknowns, and risks
- Find delay, confusion, rework, and pain
- Write a need that keeps options open
- Choose the next question to investigate

CampusConnect does not need a feature backlog yet. It needs a more truthful picture of the people, the work, and the problem.

Look Ahead: Week 3 Turns Evidence into Commitments

WEEK 2 HANDOFF

1. Stakeholders and current journey
2. Evidence, assumptions, and unknowns
3. Pain points and root-cause hypotheses
4. Needs and problem statements
5. Constraints and desired outcomes

WEEK 3 ACTION

6. Functional and quality requirements
7. Acceptance criteria
8. Priorities and risk
9. MVP and POC boundaries
10. Backlog, estimates, and delivery plan

Week 2 asks what is true enough to act on. Week 3 turns that evidence into commitments.